

# HTS Classifier

A fine-tuned 3B model that classifies trade goods on our own hardware

Technical review · [go / no-go tag-up](#)

**66–69%**

our accuracy, CLEAN-strict

**71.1%**

Gemini Flash, same exam

**0**

data leaves our environment

# A standalone classifier — no API at inference

## 1 Fine-tuned Llama 3.2 3B

Trained on CBP CROSS customs rulings (ground truth from the rulings themselves, not model guesses).

## 2 Runs on our GPU

Inference is local. No description ever leaves the environment — the argument that matters for a CBP/DHS system.

## 3 Retrieval-augmented

Candidate headings retrieved and put in the prompt; the model selects and justifies rather than recalls from memory.

## 4 Calibrated to abstain

It knows when it's unsure and defers those cases to a fallback, instead of guessing.

# Every system, one leak-free exam

Lucene — today's production

31.3%

Old pipeline (subject lines)

Fine-tuned 3B (closed-book)

3B + retrieval

Gemini Flash (frontier API)

Read this as:

2×

the current production approach

−5 pts

vs a frontier API — and we win  
on multi-item rulings

# "Why not just call Gemini?"

1

## Data control

For CBP/DHS traffic, sending every description to an external API is the hard blocker. Our model keeps all data in-environment.

2

## No per-call cost / dependency

No vendor lock-in, no rate limits, no price changes, no outage risk on the critical path. Fixed cost on hardware we own.

3

## It's already competitive

71.1% is Gemini's real score here — not the 85% folklore. We're within a few points, and ahead on the hardest (multi-item) cases.

# Where the remaining points are — and why

**The gap is architectural, not a training failure**

**4-digit heading (what ships):**

strong — a few points under Gemini.

**Full 10-digit code:**

low — a 3B can't memorize ~19,000 leaf codes. That's a coverage problem, not reasoning.

**Retrieval is the fix:**

put the candidate codes in front of the model so it selects instead of recalls.

Retrieval ceiling, measured

~93%

of items have the correct heading among the retrieved candidates (recall@20).

*The room between 66% and 93% is what select-and-justify is designed to capture.*

# Accuracy isn't the ship gate — coverage is

The system auto-accepts what it's confident about and routes the rest to a fallback. The question isn't "how often is it right," it's "how much can it safely handle alone."

**34.9%**

**of traffic auto-accepted**

*validated out-of-sample*

**78.5%**

**precision on that slice**

*above Gemini's overall accuracy*

**~55%**

**today's manual process**

*the bar we already clear*

*Business bar is 70% precision on auto-accepted items. We clear it with room, and the confident slice already beats the frontier model.*

# The measurement, not just the model

## Frozen benchmark

One 347-item test set, versioned and hashed, never trained on. Every model — ours, Gemini, Lucene — scored on it.

## Contamination engineered out

Found and removed data leakage, near-duplicate collisions, and 'answer echo' in descriptions. Early numbers were inflated; these aren't.

## One scorer for all

A single audit script grades every system. The Lucene, 3B, and Gemini numbers are directly comparable by construction.

## Honest ceilings

Where the model can't win (unwinnable items, retrieval misses) we measure and report it rather than hide it.

# The system is designed, not just the model



## Orchestration

Java + LangChain4j — fits the existing stack, no Python on the critical path.



## Confident vs. defer

Model handles the calibrated slice; ~5–10% of hard cases route to an LLM fallback.



## Drift detection

Four live signals — fallback rate, chapter distribution, embedding drift, confidence shift.



## Eval without labels

Self-consistency + invariant tests keep scoring the system in production, no new ground truth needed.

*Architecture already specced with working reference code — deployment is engineering, not research.*



# Where we are, and what's next

## Where we are

- Doubled the current production approach (31% → 66%)
- Within a few points of a frontier API, ahead on hard cases
- A validated, shippable auto-accept slice today
- Runs entirely on our hardware

## Decision for today

### Fund the next phase

Retrieval + calibration to close the Gemini gap and grow the auto-accept slice

### Move toward pilot

Stand up the designed architecture on a shadow slice of real traffic

### Bigger-model option

One-time escalation card if we need to pass the frontier outright — same dataset, larger student

*Everything built so far is model-agnostic capital — dataset, benchmark, pipeline carry forward regardless of the path chosen.*